



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

Project documentation

Course Title: Mobile Computing

Course Code: CSE 4203

Project name: Medical Report Analyzer

Submitted To:

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Submitted By:

Team : 02

Section: 8D

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Team Information

Team Name: CSE4204-8D-T02

Section: 8D

Project Title: Medical Report Analyzer

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Github link : <https://github.com/Tushin-vai/CSE-4204-8D-T02>

Project Information

Project Overview

The Medical Report Analyzer is an AI-powered web application designed to bridge the gap between complex medical documentation and patient comprehension. By leveraging large language models, the system translates clinical terminology, lab values, diagnoses, and prescriptions into plain, readable language that any patient can understand without a medical background.

Objectives

- Provide patients with instant, plain-language summaries of their medical reports.
- Extract and highlight key findings such as diagnoses, lab values, and abnormal results.
- Offer an interactive chatbot for follow-up questions grounded in the uploaded report.
- Securely store reports, summaries, and chat histories per authenticated user account.

Support both text-pasted and image/scanned report uploads via OpenAI Vision

Core Features of the System

1. Medical Report Upload & Input

Users can upload medical reports in multiple formats, including text, PDF, and scanned images. The system processes and prepares these inputs for analysis.

2. AI-Powered Report Analysis

The system leverages AI to interpret medical reports and provide:

- Plain-language summaries for easier understanding
- Extraction of key findings such as diagnoses, lab values, and medications
- Identification and highlighting of abnormal or out-of-range results

3. Interactive Medical Chatbot

Users can engage in a context-aware chatbot that answers follow-up questions based on their specific medical report, enabling deeper understanding and clarification.

4. User Authentication & Account Management

Secure authentication is implemented using JWT, supporting:

- User registration
- Login and logout functionality
- Session management and protection of user data

5. Report History Management

Users can access, review, and manage all previously uploaded and analyzed medical reports within their account dashboard.

6. Data Storage & Security

All user data, including reports, summaries, and chat history, is securely stored using Supabase with strict access control policies to ensure privacy and data protection.

System Architecture Design

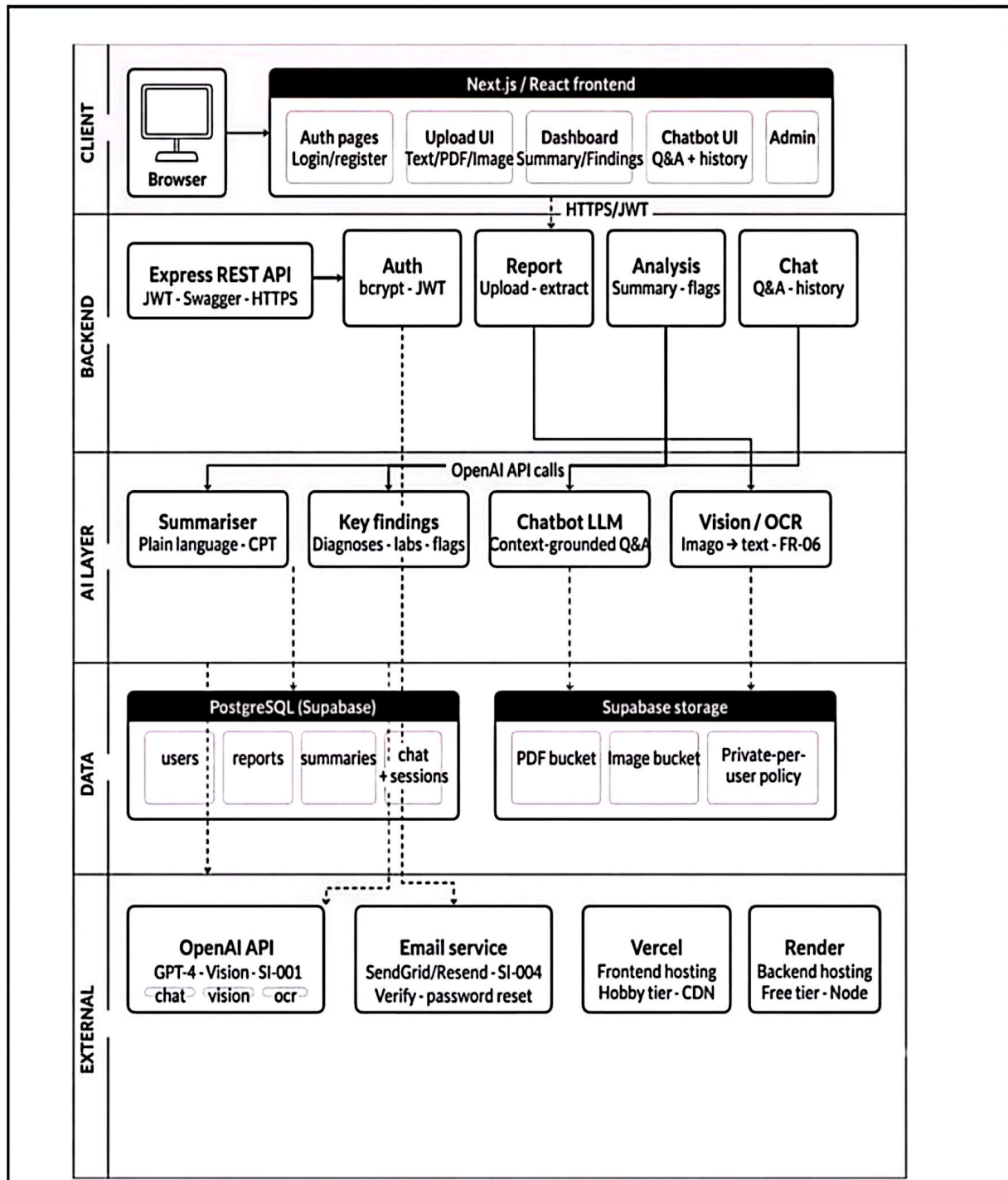


Fig 1 : System Architecture Design

Entity Relationship (ER) Diagrams

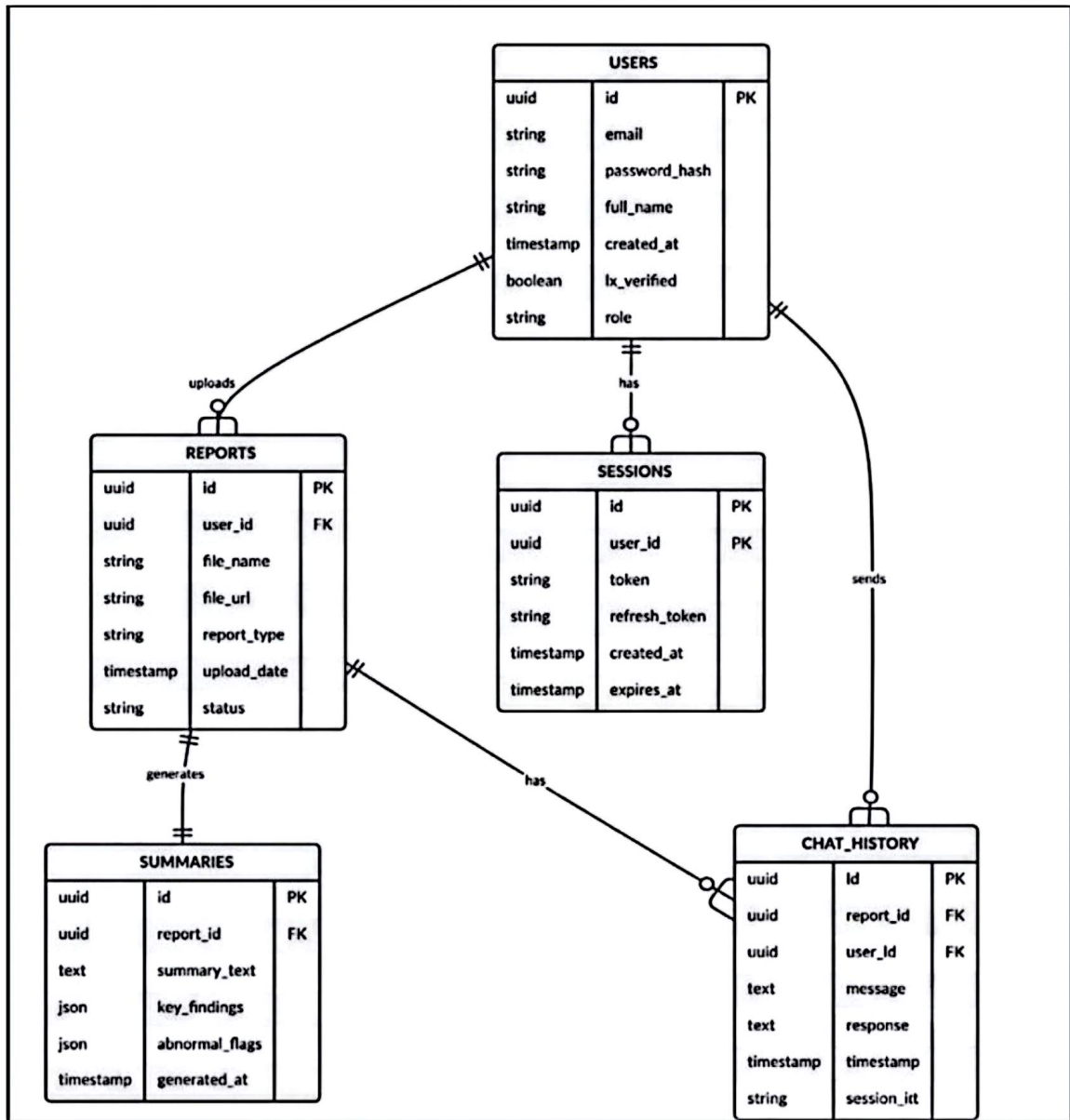


Fig 2 : Entity Relationship (ER) Diagrams

Use case Diagram

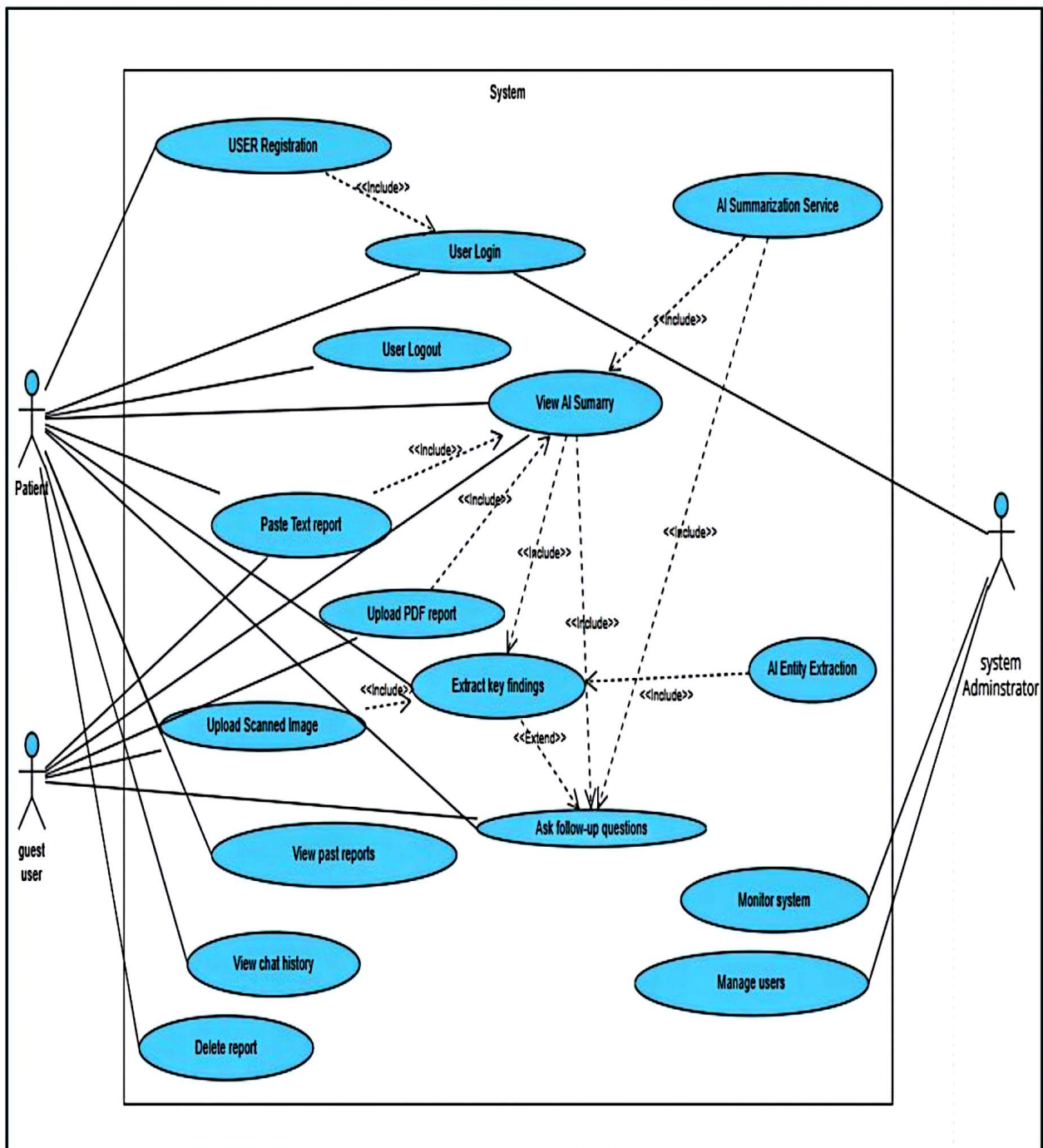


Fig 3 : Use case Diagram

Activity Diagram

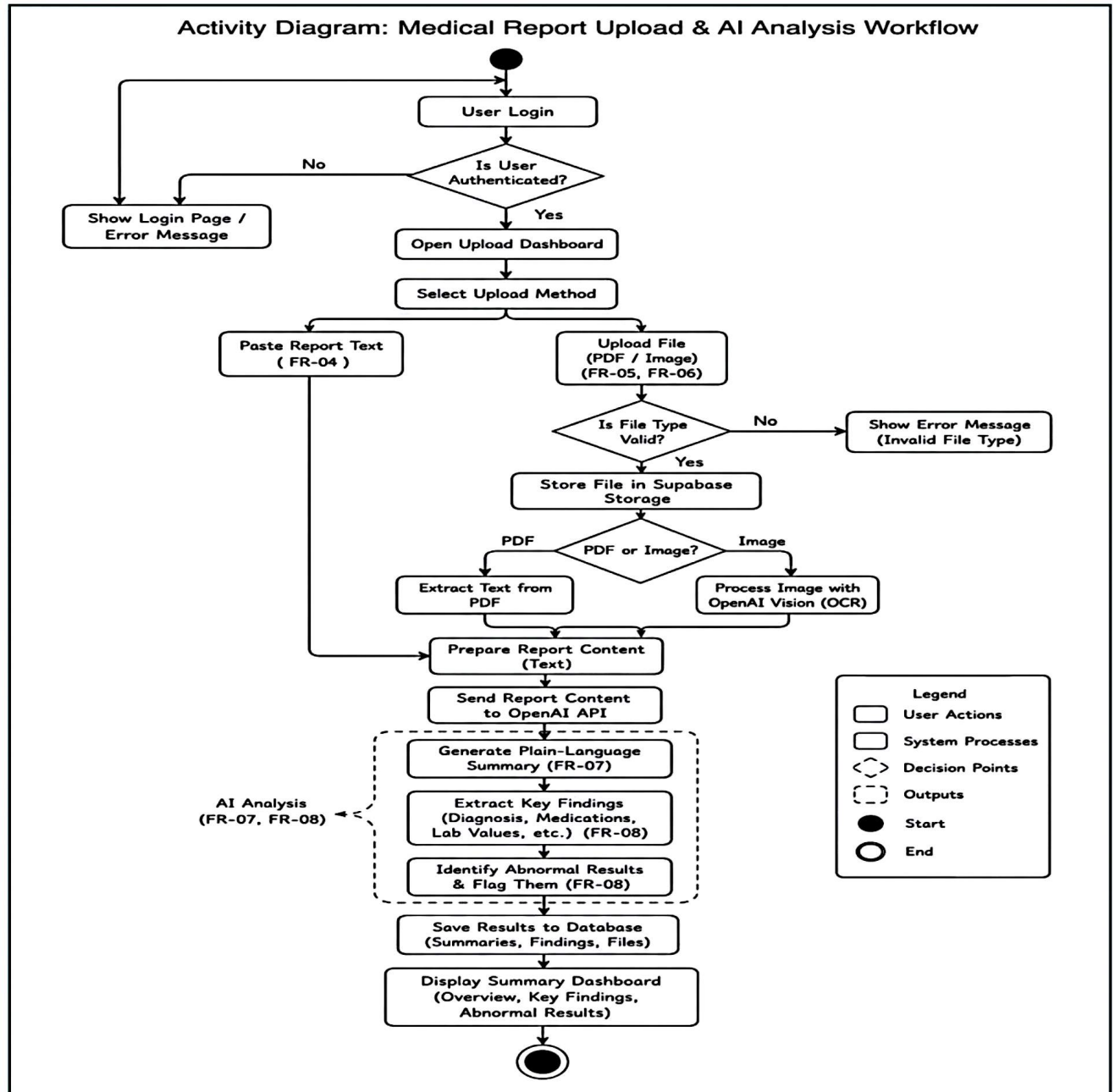


Fig 4 : Activity Diagram

API Design Document

Authentication APIs

1. POST /register

Purpose:

Register a new user account in the system.

Input Data:

```
{  
  "full_name": "John Doe",  
  "email": "john@gmail.com",  
  "password": "123456"  
}
```

Expected Output:

```
{  
  "message": "User registered successfully",  
  "user_id": 1  
}
```

2. POST /login

Purpose:

Authenticate user and generate JWT token for session management.

Input Data:

```
{  
  "email": "john@gmail.com",  
  "password": "123456"  
}
```

Expected Output:

```
{  
  "message": "Login successful",  
  "token": "jwt_token_here"  
}
```

3. POST /logout

Purpose:

Invalidate user session and log out the user.

Input Data:

```
{  
  "token": "jwt_token_here"  
}
```

Expected Output:

```
{  
  "message": "Logout successful"  
}
```

User APIs

4. GET /profile

Purpose:

Fetch logged-in user profile information.

Input Data:

- Authorization Header: JWT Token

Expected Output:

```
{  
  "id": 1,  
  "full_name": "John Doe",  
  "email": "john@gmail.com",  
  
  "role": "patient"  
}
```

5. PUT /profile

Purpose:

Update user profile information.

Input Data:

```
{  
  "full_name": "John Updated"  
}
```

Expected Output:

```
{  
  "message": "Profile updated successfully"  
}
```

Report APIs

1. POST /report/upload

Purpose:

Upload a medical report (text, PDF, or image) for AI analysis.

Input Data:

- File (PDF/Image) OR:

```
{  
  "text": "Patient medical report content..."  
}
```

Expected Output:

```
{  
  "message": "Report uploaded successfully",  
  "report_id": 101,  
  "status": "processing"  
}
```

2. GET /report/:id

Purpose:

Retrieve a specific report with AI summary and findings.

Input Data:

Report_id (URL parameter)

Expected Output:

```
{
  "report_id": 101,
  "summary": "Patient has mild anemia...",
  "key_findings": {
    "diagnosis": ["Anemia"],
    "abnormal_values": ["Hemoglobin low"]
  }
}
```

3. GET /reports**Purpose:**

Get all reports of a user.

Input Data:

- Authorization token

Expected Output:

```
{
  "reports": [
    {
      "report_id": 101,
      "uploaded_at": "2026-06-15",
      "summary_preview": "Mild anemia detected..."
    }
  ]
}
```

4. DELETE /report/:id**Purpose:**

Delete a specific medical report and all related data.

Input Data:

- report_id (URL parameter)

Expected Output:

```
{
  "message": "Report deleted successfully"
}
```

AI Chat APIs

1. POST /chat

Purpose:

Ask questions about a specific medical report using an AI chatbot.

Input Data:

```
{
  "report_id": 101,
  "message": "What does low hemoglobin mean?"
}
```

Expected Output:

```
{
  "response": "Low hemoglobin means your blood has fewer red blood cells than normal..."
}
```

2. GET /chat/history/:report_id

Purpose:

Retrieve chat history for a specific report.

Input Data:

- report_id (URL parameter)

Expected Output:

```
{
  "history": [
    {
      "message": "What is anemia?",
      "response": "Anemia is a condition where..."
    }
  ]
}
```

AI Integration Workflow

Why AI is needed :

AI is required in this system to analyze complex medical reports and convert raw medical data into simple, understandable insights for users. Most users cannot easily understand medical terms, lab results, or diagnostic summaries. AI helps by:

- Translating medical jargon into simple language
- Summarizing long medical reports
- Detecting possible health issues from report data
- Answering patient questions in real-time
- Providing personalized explanations based on report context

Without AI, users would need a doctor to interpret every report manually, which is slow and costly.

Which AI service/model will be used

The system can use one of the following AI models:

- **OpenAI API or Google Gemini API**

These models are suitable because they:

- Understand medical text well
- Generate human-like explanations
- Support context-based chat (important for reports)

Input to the AI system

AI receives structured and unstructured data from backend APIs.

Input types:

1. Medical report text (OCR or uploaded text)

Patient has Hb level 9.5 g/dL, symptoms include fatigue and dizziness.

2. User chat question

What does low hemoglobin mean?

3. Combined context input:

```
{  
  "report_summary": "Mild anemia detected...",  
  "key_findings": ["Low hemoglobin", "Fatigue"],  
  "user_question": "Is this dangerous?"  
}
```

Output from the AI system

AI returns structured medical explanations.

Example output:

```
{  
  "response": "Low hemoglobin means your blood has fewer red blood cells than normal. This condition is called anemia and may cause fatigue, weakness, and dizziness. It is usually treatable depending on severity."  
}
```

How AI improves the project

AI improves the system in several important ways:

- **Better user understanding:** Converts complex medical reports into simple language
- **Time saving:** No need for doctor interpretation for basic understanding
- **Interactive experience:** Users can ask questions anytime
- **Personalized responses:** Answers are based on the user's own report
- **Scalability:** One AI system can handle thousands of users simultaneously
- **Smart insights:** Helps highlight abnormal values and possible issues.

AI Workflow diagram

